

Converting heterogeneous complex geological models to consistent finite element models: methods, development, and application to deep geothermal reservoir operation

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Abstract Static geological models representing complex geological systems are the prerequisite of dynamic model simulations applied for assessing subsurface processes. The corner point grid approach has been applied to represent the complexity in geometry, hydraulic connectivity, and heterogeneity found in these static geological models. Due to the occurrence of faults, pinch-outs, and eroded geological layers, corner point grids easily degenerate, which leads to model inconsistencies. This study introduces a workflow for converting heterogeneous geological models to consistent finite element models, accounting for regular and irregular hexahedral blocks of the corner point grid by converting to a set of hexahedra, prism, pyramid, and tetrahedral elements, based on the individual degeneration situation. Heterogeneous geological data such as permeability or porosity can be transferred layer-wise or on a block-wise basis. Additionally, well trajectories can be accurately mapped to the converted finite element mesh, to place the corresponding source terms. The developed workflow is tested on dedicated test cases and applied to convert a real complex field site from the North German Basin for use in a deep geothermal reservoir operation. The field application demonstrates the robustness and applicability of the newly developed conversion workflow and the suitability of the converted mesh for dynamic finite element reservoir model simulations.

Keywords Static models · Dynamic models · Corner point grid · Finite element mesh · Geothermal reservoir operation · ANGUS+ · [OpenGeoSys](#)

Introduction

Mathematical modeling of subsurface processes in geoscientific applications requires adequate representation of both the static subsurface as well as the process dynamics. Applications include geothermal energy extraction, ground-water abstraction, oil and gas recovery, underground storage of natural gas as well as deposition of hazardous materials in the subsurface. In recent years, due to the ongoing transition of energy supply in Germany toward an increased use of energy from renewable sources like wind or solar power, the geological subsurface is investigated as a georeservoir for renewable energy storage, e.g., via geothermal heat storage or gas storage in caverns or porous formations (Bauer et al. 2013). To investigate the use of the geological subsurface and to quantify possible induced impacts, a multidisciplinary approach is required, which combines fields of geology, numerical modeling, geochemistry, geophysics, etc. (Blöcher et al. 2010b; Franco and Vaccaro 2014; Sa et al. 2014; Bauer et al. 2015; Pfeiffer and Bauer 2015). Within this approach, a static model, which can correctly represent the geological nature of the studied reservoir and its environment in the subsurface, is the prerequisite in order to achieve a reliable evaluation and assessment through numerical simulations. Dynamic models are thus required for simulating reservoir behaviors and dimensioning of the individual operation as wells as for the assessment of induced effects.

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Large-scale geological models in the subsurface often show a high complexity in geometry and hydraulic connectivity, due to the existence of faults, pinch-outs, and eroded layers. With the interest of exploring and exploiting hydrocarbon reservoirs in the subsurface, the petroleum industry has invested large efforts in building static geological models and has developed powerful geological reservoir modeling codes, such as GOCAD (Geological Object Computer Aided Design) (Paradigm 2016) or Petrel (Petrel E&P Software Platform) (Schlumberger 2014a). Two basic approaches, i.e., a surface-based and a volume-based approach, are applied to model the geometrical complexity of the subsurface. The surface-based approach (Neave 2007) is commonly known as the pillar-gridding technology, which firstly models faults as pillars and horizons as smooth planes, and later creates geological layers by following defined topological relations. The volume-based approach by Souche et al. (2013) models geological layers directly by using an unstructured mesh which is internally constrained by faults. Afterward, the resulting geological models need to be discretized for simulation purpose, and the corner point grid (Ponting 1992) has been utilized in the above-mentioned commercial software as an industrial standard (Aarnes et al. 2008). The corner point grid is represented by a set of hexahedral blocks with eight nodes and 12 edges. The length of vertical or inclined edges stands for the thickness of the formations, and the eight corner points represent the top and bottom horizon depths of geological formations. In complex subsurface structures, the thickness of the formations becomes smaller or even zero where the formations are partially or completely eroded and where pinch-outs exist. This leads to reduced or zero length of those vertical or inclined edges, and thus degenerated or even disappearing blocks.

To transfer a corner point grid-based geological model to a workable dynamic reservoir model, a procedure of grid and property data conversion is required. Generally, two approaches exist to convert models to be used in grid-based numerical schemes, as, e.g., finite volume (FV), finite difference (FD), or finite element (FE) methods. In the first approach, the corner points of important entities, e.g., faults and horizons, are extracted from the geological model, and the numerical mesh is constructed using existing meshing tools [e.g., the 'MeshIt' tool developed by Cacace and Blöcher (2015)] or through a resampling procedure [e.g., the 'GO2OGS' developed by Fischer et al. (2015)]. This approach is similar to the volume-based approach in geological modeling methods, and it can have many freedoms in modeling subsurface geological complexity, e.g., faults and fractures, by a consistent unstructured mesh (e.g., Blöcher et al. 2010a; Zehner et al. 2015). In this approach, the parameters can be consistently transferred to the

elements based on their geological group index, but within a geological group, the discrepancy will be introduced element-wise in transferring subsurface heterogeneity parameterized on an elemental level in the geological models to reservoir models, e.g., permeability and porosity, and thus, the generated mesh is often parameterized after the conversion before being utilized in reservoir simulations. The approach also needs substantial technical steps and procedures, to arrive at a consistent simulation mesh. The second approach utilizes corner point grid-based geological models, which are fully discretized and parameterized for simulation purpose. It directly converts individual blocks in the corner point grid by maintaining their individual polyhedral geometry, and transfers heterogeneous parameters from individual blocks to converted elements. Although the representation of geological complexities in the converted mesh is restricted by their representation in the original corner point grid, this approach produces consistent reservoir models without reconstructing the mesh and reparametrizing it afterward. This is thus a comparatively fast approach.

The direct conversion approach has been implemented by Ni and Chen (2014) for GOCAD SGrid models, and by Graupner et al. (2011) and Park et al. (2014) for Petrel grids. These implementations have the restriction that the corner point grid does not contain any degenerated blocks. However, in the presence of degenerated blocks, as they necessarily occur when more complex geological features are included, inconsistent elements arise, and thus, the conversion from blocks to only hexahedral elements fails. To resolve this, Aarnes et al. (2008) implemented a non-trivial scheme by partitioning each block into six or less tetrahedrons. Since a geological model often contains millions of blocks in a grid, this scheme will increase the number of corresponding elements by almost a factor of six after conversion, and thus lead to large computation times of the resulting reservoir simulation models. Accordingly, besides hexahedron and tetrahedron, the direct conversion of degenerated blocks of a corner point grid requires the implementation of other types of polyhedrons. During the conversion as well as in the FE code, only this partitioning approach allows for a geometrically consistent representation of the geological features in the numerical mesh and ensures a consistent computational mesh for the process simulation. As the elements can inherit parameters and data straightforwardly from the partitioned blocks, this direct conversion approach also guarantees a consistent parameter and data transfer between the corner point grid and the FE mesh. This assists to couple commercial reservoir simulators with FE-based open source software (Graupner et al. 2011; Pfeiffer et al. 2016), e.g., the coupling between the Eclipse reservoir simulation software (Schlumberger 2014b) with OpenGeoSys (Kolditz and Bauer 2004) and

can be especially used to simulate induced geochemical reactions (e.g., Mitiku et al. 2013; Li et al. 2014).

In addition to a consistent numerical mesh representing subsurface features in reservoir models, wells are the main elements of an injection and extraction system. Open well screens describe the locations of interaction with the geological formations, e.g., locations for water and gas injection and extraction, and thus locations of largest pressure changes and highest occurring gas saturation (Benisch and Bauer 2013; Benisch et al. 2014). Besides, wellbores penetrating through geological formations are widely considered as the main possible leakage pathways, e.g., production or abandoned wells in geologic carbon sequestration (Pan et al. 2011). Due to the geometrical complexity and the scale of dynamic reservoir models, wells are often first designed in static geological models and later transferred to dynamic models. For instance, wells designed in Petrel can be exported as connection and completion data arrays to the Eclipse reservoir simulation software. Among the first-kind approaches of converting geological models to dynamic models, 'MeshIt' from Cacace and Blöcher (2015) is able to adapt well data as 1D polylines into an FE mesh. However, to our knowledge, none of the direct conversion approaches have the same feature.

In this study, therefore, we introduce a new workflow to convert a corner point grid-based static model to a consistent finite element-based dynamic model. This workflow avoids the problems of inconsistent data transfer or excessive increase in element number by an efficient and accurate grid block partitioning strategy, which honors the individual geometrical situations of each block. Geological parameters, such as permeability and porosity, are transferred from the corner point grid blocks to the FE mesh elements accounting for block partitioning. Additionally, well trajectories, open screen lengths, and corresponding well data are transferred as nodes and polylines in the converted FE mesh. As a proof of concept, we implement the new workflow into a mesh and parameter converter, and test the software by applying it to model a hypothetical geothermal reservoir operation at a real geological site in Schleswig-Holstein, northern Germany.

Corner point grid conversion workflow

Blocks and preprocessing

In a corner point grid, a set of hexahedral blocks with eight corner nodes and 12 edges is able to represent the subsurface complex geological structures, such as faults, pinch-outs, and eroded layers, but not the subgrid fracture networks, which are often modeled as grid properties. In

individual blocks, each set of four nodes at the top and bottom follows the spatial geometric shape of formation horizons, and the four vertical or inclined edges in between stand for the thickness of formation layers.

Figure 1a shows a side view in the x - z plane of a corner point grid containing a fault structure. At the fault plane (see Fig. 1a closer view), the nodes of the blocks representing the formation layers adapt to the fault slip, which creates non-neighbor blocks at different sides of the fault plane. The length of vertical or inclined edges changes with the varying thickness of the formation layers close to the fault. Here, the formation layers are shifted and not eroded completely, so the formation thickness is not zero, and thus, the blocks remain as eight-node hexahedral blocks. Figure 1b shows a side view in the x - z plane of a corner point grid representing a stack of geological formations containing anticlinal structures. In this side view, the thickness of the geological formations varies significantly due to erosion. Correspondingly, the length of those four edges in each block changes as well (see Fig. 1 closer view). When a formation becomes thinner, those four edges become shorter. When a geological formation is completely eroded, i.e., its thickness reduced to zero, those four edges in the blocks will also reduce to zero length. This can be clearly seen in Fig. 1, e.g., for the light green layer.

For a block, any number of those four edges can reduce to zero length, i.e., collapse, and thus, a variety of degenerated block types exists. A 3D view of possible types of degenerated blocks in a corner point grid representing a typical eroded formation layer is shown in Fig. 2. The degenerated blocks often appear close to where erosion happened, e.g., close to the fold center of an anticline. The blocks marked with red lines represent the eroded part of the formation, i.e., blocks with four vertical edges of length zero, and thus also zero volume. In the transition area, different numbers of those four edges collapse and form degenerated five-node, six-node, and seven-node blocks, respectively. The remaining part of the layer is conformed and consists of eight-node hexahedral blocks.

Due to the problems caused by degenerated blocks, e.g., zero-volume blocks, double counting nodes, and wrong neighboring information, a preprocessing on the block data structure is performed before executing the individual block partitioning procedure. In corner point grids, zero-volume blocks disappear and are marked as inactive blocks. While the grid is read into the mesh converter, we thus remove those blocks directly and only store active blocks with nonzero-volume ones as eight-node structure as they appear in corner point grids. Due to the existence of five-, six-, and seven-node blocks, more than one node appears at one position after the edges collapse and this leads to non-unique node numbering. Therefore, a

Fig. 1 Side view of a corner point grid with *vertical pillars* including faults (a) and anticlinal structures (b). Different colors indicate different formations, designated by uniform material indices

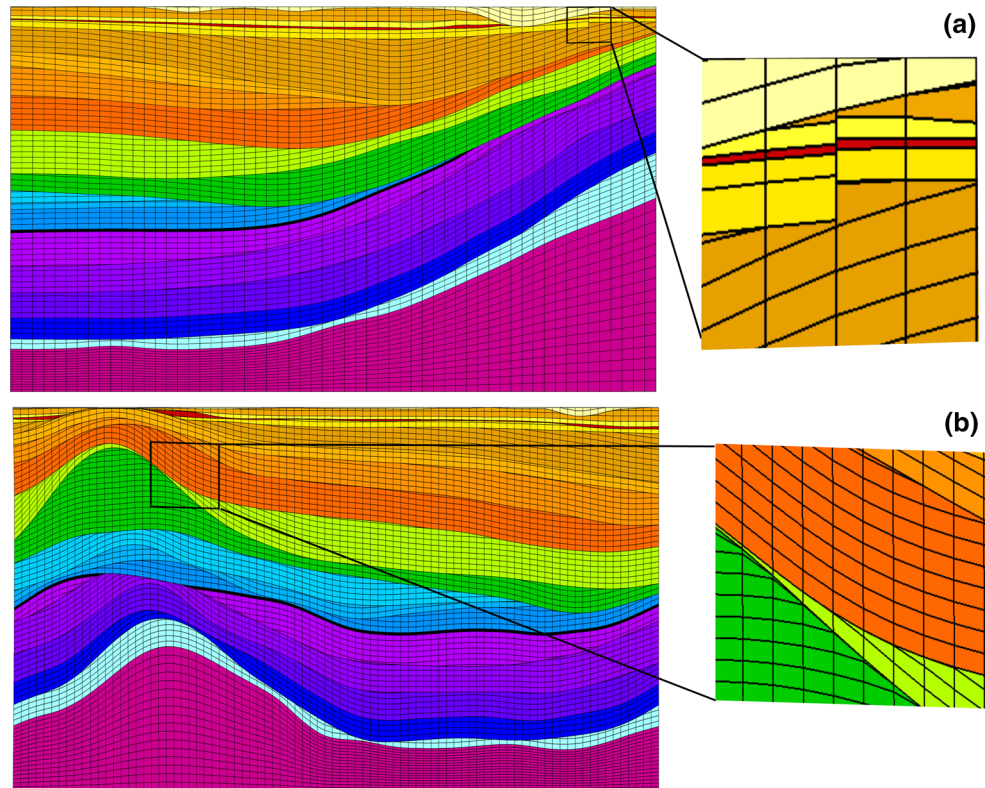
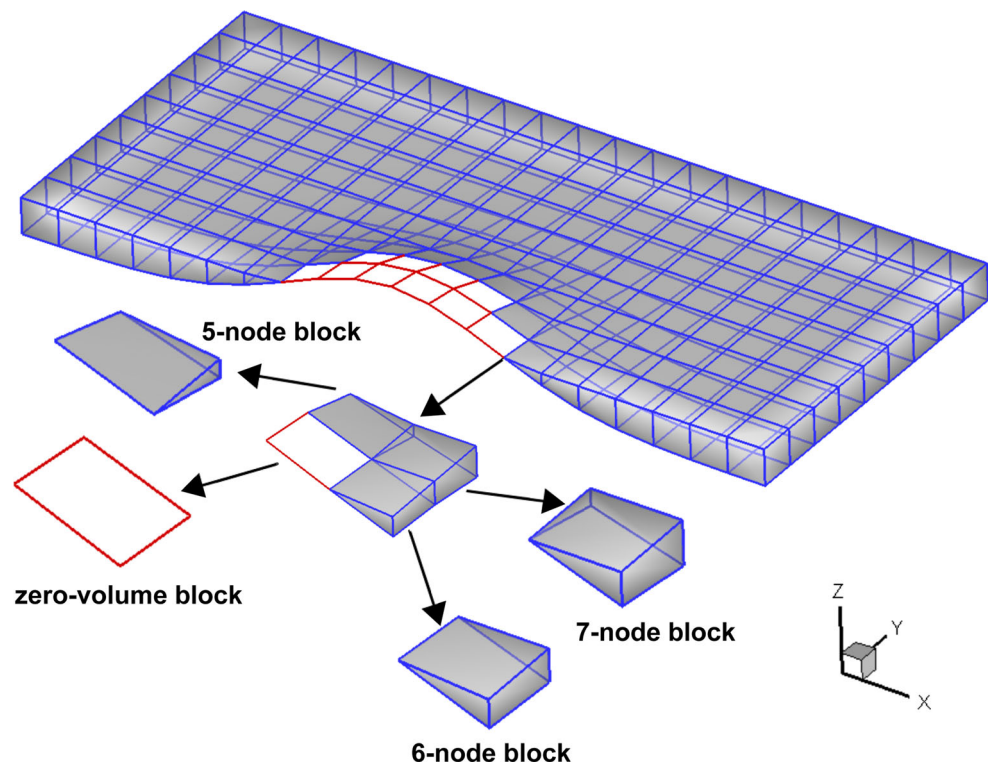


Fig. 2 A 3D view of possible types of degenerated blocks. Blocks with *red lines* represent zero-volume blocks; blocks with *blue lines* represent degenerated blocks and hexahedron blocks



procedure for numbering the nodes uniquely loops over all the blocks, compares the coordinates of each node, and removes the repetitive ones. A node list consisting of

individual unique indices and the node coordinates is constructed, and these unique indices are assigned back to the nodes of corresponding blocks in their eight-node

structure. Connectivity information, e.g., faces and neighbors, is adapted to account for zero-volume and degenerated blocks. Some blocks have less than 12 edges, three-node faces, and non-neighbor connections. On the basis of the unique node numbering, the faces of each block are reconstructed, and each face is a unique entry containing node indices, number of face nodes, face normal vector, and block indices this face belongs to. Similar to the node, the face indices are assigned back to each block, and thus, neighboring information between blocks can be determined by comparing their joint faces. Two blocks with zero-volume blocks in between are marked as non-neighbor connection in corner point grids, but in FE meshes, we mark them as neighbors because they share the same face. This is of great importance to the following partitioning conversion scheme because neighbor blocks have to maintain consistent joint faces in FE meshes.

Individual block conversion

As explained in section ‘[Blocks and preprocessing](#)’, the degeneration is caused by collapsed vertical or inclined edges in each block. To handle the degenerated blocks (see Fig. 2) and consistently convert them to finite elements, these blocks are partitioned according to the newly developed workflow. How each block is partitioned is determined by the specific type of degenerated block. According to the number of collapsed edges, five block types exist, and the conversion scheme of each type is schematically illustrated in Fig. 3. Only one representative case for each type is shown, but the converter accounts for all variations in each type due to changes in the positions of collapsed edges.

- Type 1 None of the four vertical edges of this block has zero length. The block is converted to a hexahedron element. However, if in the same column, the neighbor block above or below is partitioned, then this hexahedron element is also partitioned into two prism elements (red dashed lines) in order to maintain consistent joint faces between the elements.
- Type 2 One of the four vertical edges of the block has zero length. The block is divided into one prism element and one pyramid element.
- Type 3a Two neighboring vertical edges at the same face of the block have zero length. The block is converted to a prism element. Similar to type 1, if in the same column, the neighbor block above or below needs to be partitioned, then this prism element is further partitioned into 1 pyramid element and 1 tetrahedron element.

- Type 3b Two opposing vertical edges at opposite faces of the block have zero length. The block is converted to two pyramid elements.
- Type 4 Three vertical edges of the block have zero length. The block is partitioned into two tetrahedron elements. If one of the tetrahedrons is formed from the three zero-length edges, it has a volume of zero and thus is removed from the mesh.
- Type 5 Four vertical edges of the block have zero length. The block has a volume of zero and is removed from the mesh.

Through the above approach, a corner point grid will be consistently converted to a hybrid FE mesh with multi-type elements, i.e., tetrahedrons, prisms, pyramids, and hexahedrons. In case of faulting, the blocks at two sides of a fault will be converted each to hexahedral elements following Type 1. The degenerated blocks at pinch-outs and erosional layers will be converted to tetrahedral, prism, and pyramid elements following Type 2–5, depending on the exact geometrical setting for each block. The outlined scheme has been implemented into a stand-alone mesh converter, which can be operated under all operating systems. As an additional option, the converting scheme by Aarnes et al. (2008), i.e., partitioning each block into six or less tetrahedrons, has been included as well.

Parameter conversion

To parameterize a corner point grid using the property modeling approach (Schlumberger 2014a), each geological formation can be considered as individual material, and thus, a uniform material index is assigned (see Fig. 1 where each color indicates a uniform material index). Additionally, based on geostatistical or stochastic methods, heterogeneous property data can be assigned to each block in each formation according to its sedimentation features. The parameter data are stored block-wise for the corner point grid, and therefore, the data are transformed block-wise from corner point grids to FE meshes. As shown in Fig. 3, the green dots represent the data being transferred. Different parameters are stored as different entities in each individual block. Through block conversion, the elements will inherit the data directly from the blocks from which they are partitioned. The geological group index is always transferred from blocks to elements in order to identify the geological formation the elements belong to. Within a geological formation, i.e., geological group, if heterogeneously distributed parameters exist, they are also transferred from individual blocks to elements.

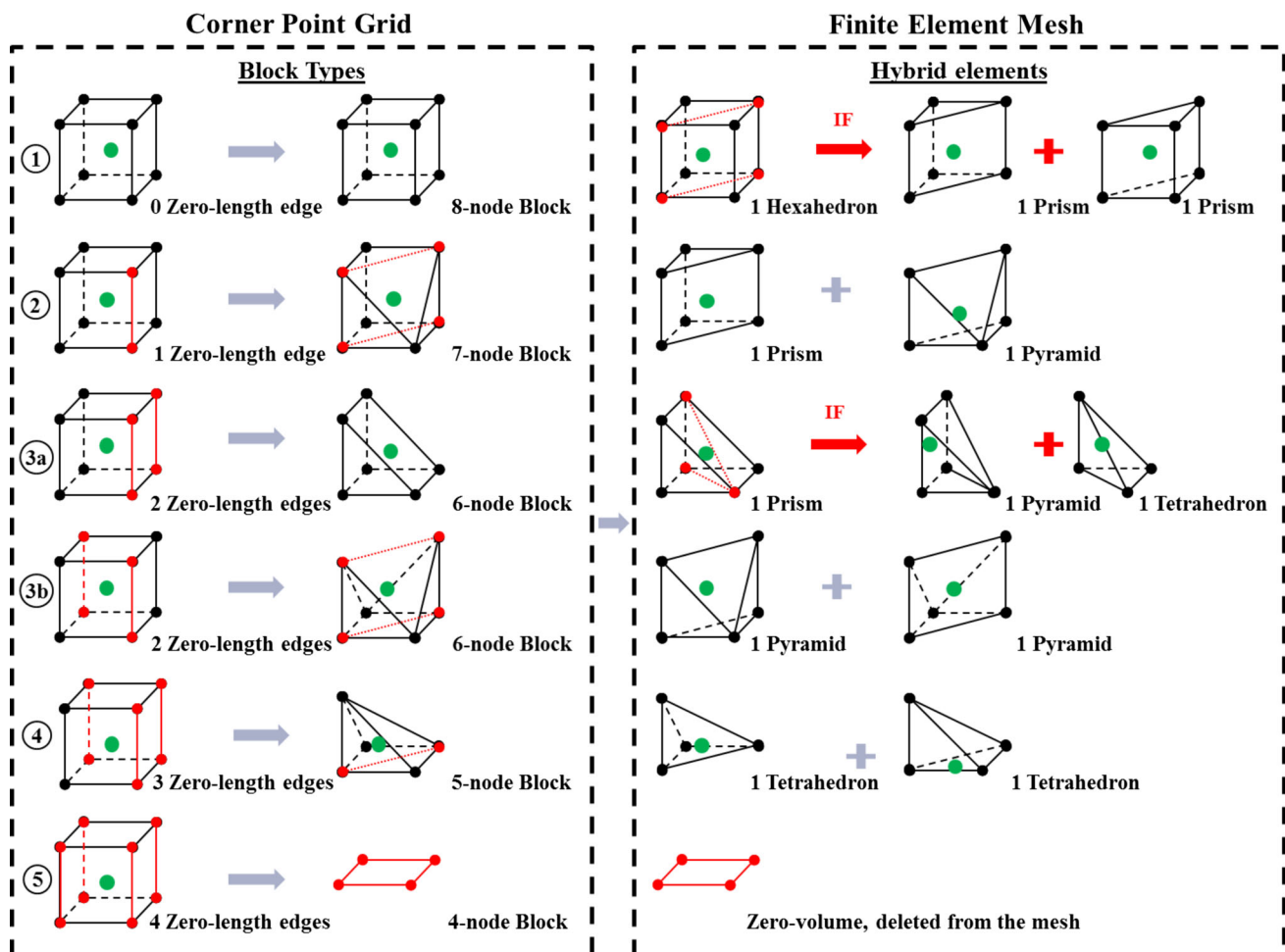


Fig. 3 Converting schemes between a corner point grid and a FE mesh. Red solid lines represent collapsed edges (zero length) in degenerated blocks; red dashed lines represent partitioning schemes;

green dots illustrate transferred property data. The partitioning marked 'IF' is only performed if neighboring blocks are partitioned

Well trajectory conversion

Generally, a well trajectory consists of well head, well path, and well screen. A set of well head coordinates indicates where the well starts, and a list of block data shows where the well penetrates the grid. In an FE mesh, wells can be represented by a list of nodes (0D geometric feature) or a polyline (1D geometric feature) (Cacace and Blöcher 2015), which are used as locations for source points or boundary conditions for the injection or extraction sections. To make use of well trajectories defined within a corner point grid, a procedure of transferring blocks penetrated by well trajectories to nodes in the FE mesh is required. An approach to identify well nodes in the FE mesh by determining the closest node along the well trajectory is implemented. The well trajectory in Petrel lists the indices of penetrated blocks. It also includes the information on how the well penetrates blocks by listing the intersection points on two block faces, i.e., one entry

face and one exit face for each block. On each face, therefore, we evaluate the distances from the intersection point to the corner nodes attached to this face and then apply the following approach to determine which node should be picked to represent the well position (Fig. 4). These cases are depicted in Fig. 4:

- Case 1 On each face, the node with the smallest distance to the intersection point is identified and picked.
- Case 2 On one face, the closest node to the intersection point is identified and picked. If, however, on the other face of the same element, more than one node has the same distance to the intersection point, then the distances between the already picked node on the first face and the nodes with identical distances to the intersection point on the other face are calculated, and the node with the closest distance is picked. This procedure ensures that the well path is mapped along vertical edges of blocks as often as possible.

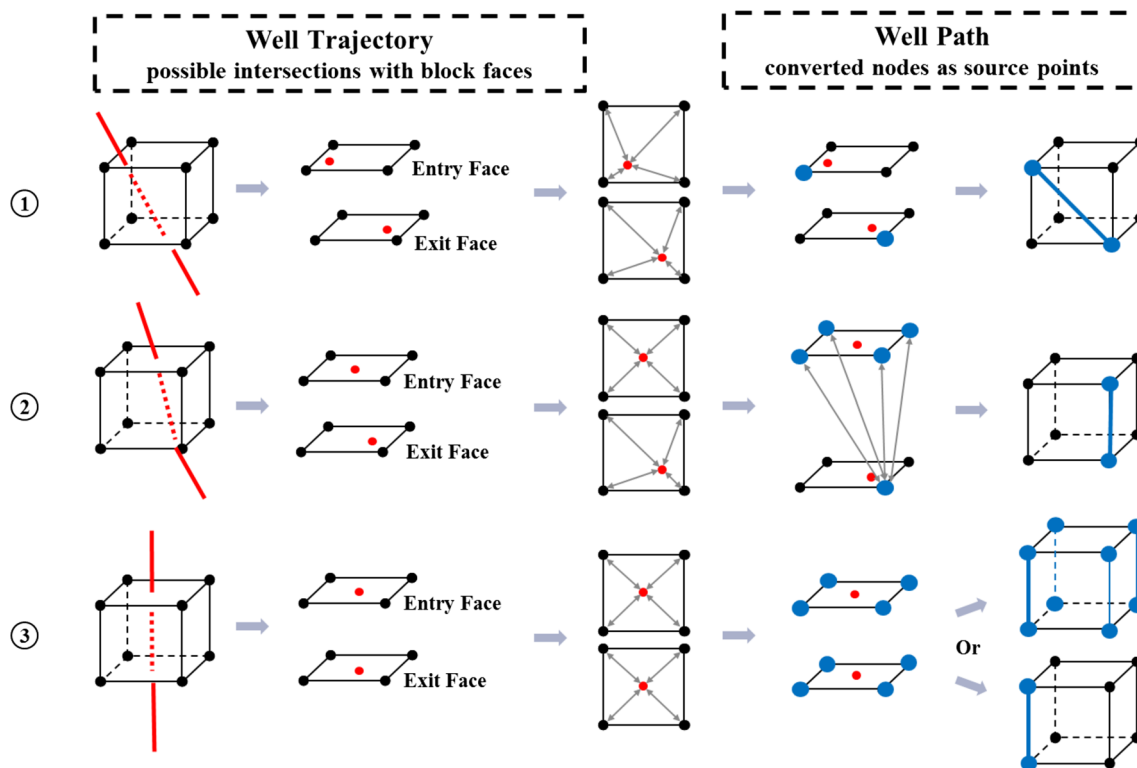


Fig. 4 Converting schemes of well trajectory. Red lines: well trajectory penetrating blocks; Red dots: intersecting points on the entry and exit faces; Gray arrow lines: distance between the

intersecting point to nodes or between two nodes; Blue dots: picked nodes; Blue lines: converted well path

Case 3 On both faces, more than one node has the same distance to the intersection point. This case happens only in geometrically simple settings of synthetic models. In this case, all the nodes with the same distance on the face are picked and each set of two closest nodes between the two faces is considered as a well path. This yields more than one consistent well path for the FE mesh. Either one or more than one of them can be used to locate the well source terms of the dynamic model.

The open-hole section along a well trajectory needs to be modeled as grid property to indicate which penetrated blocks are connected to the open-hole section. The output file of the well trajectory lists this information for each penetrated block, and thus, the nodes belonging to those blocks to represent the open-hole section of the designed well trajectory will be picked as described above.

After picking the nodes for the well path and the well open-hole section, the well trajectory is represented as a polyline, i.e., as 1D geometry. The geometrical features of the converted well path and the open-hole sections are written as output for the use of source term or boundary condition placement in the FE-based simulators.

Used programs

Petrel

Petrel (version 2014.6) is a commercial software platform providing integrated workflow tools to build static geological reservoir models later used in dynamic reservoir modeling simulators as, e.g., ECLIPSE (Schlumberger 2014a, b). The corner point gridding approach is one of the methods in Petrel to construct geological structural models. The approach provides several output files regarding the corner point grid. The ‘*.FGRID’ file contains formatted grid data, e.g., grid dimension, coordinates of corner points, non-neighbor connections, etc. The ‘*.INC’ file contains block-wise parameter data. The well trajectory data can be extracted from the well connection file ‘*.WCND’.

OpenGeoSys

OpenGeoSys (OGS) is an open source, multi-platform finite element-based scientific modeling software which can simulate individual or coupled thermo-hydro-mechanical-chemical (THMC) processes in porous and fractured media (Kolditz and Bauer 2004). OpenGeoSys (www.opengeosys.org).

opengeosys.org) has been successfully applied to reservoir simulations concerning geothermal heat extraction (Watanabe et al. 2009; Boockmeyer and Bauer 2014), CO₂ storage (Li et al. 2014; Park et al. 2014), and density-driven flow. To perform those reservoir simulations, OGS needs usable FE meshes generated or transferred from other platforms (Kalbacher et al. 2005, 2007; Miles et al. 2007). OGS provides all finite element types utilized in the mesh converter, i.e., hexahedron, prism, pyramid, and tetrahedron elements. The '*.msh' file, i.e., FE mesh file, contains a list of nodes with coordinates and a list of elements indicating element type, geological group index, and corresponding node topology. The '*.gli' file contains source point nodes indicating well trajectories, and the '*.property' file contains the element-wise property (one file for each property).

Mesh converter

To transfer from static geological models to FE-based dynamic models, the newly implemented mesh converter offers the workflow of converting a corner point grid, parameter data, and well trajectories to an FE mesh, element properties, and source points. The mesh converter is a stand-alone program, which is written in the object-oriented C++ programming language. As a proof of concept, all the read-in and write-out functions are developed referring to the format of Petrel (version 2014.6) output files and input files required by the OGS simulator. Although those functions are software related, only simple modifications are required to apply the mesh converter to other corner point grids-based software and FE-based simulators. Additionally, in order to support the coupling between OGS and the ECLIPSE reservoir simulator implemented by Graupner et al. (2011) and Pfeiffer et al. (2016), the mesh converter generates files to list block indices and connectivity information. We have tested and verified the mesh converter by a series of artificial dedicated but very technical benchmarks and test cases, e.g., geological models containing degenerated blocks like the one shown in Fig. 2. The benchmarks and test cases are very technical in nature and therefore not reported here in detail, but the developed methods have proven robust in all these benchmarks and test cases. In the following, we present an application to model a hypothetical geothermal reservoir operation at a real geological site in Schleswig–Holstein, northern Germany.

An application to a deep geothermal reservoir operation

To make use of and test the newly developed workflow, a hypothetical scenario of a deep geothermal reservoir operation is numerically simulated at a real geological site in

Schleswig–Holstein, northern Germany. We apply a well doublet in a high-permeability saline formation of the middle Rhaetian at about 4000 m depth, in which the saline water has a temperature of about 160 °C. The well doublet is assured to operate for 6 months a year, and reservoir behavior is simulated for 20 years. During the operation, hot water is extracted at the 'warm' extraction well at a rate of 150 m³/h, while the cooled water is re-injected at the 'cold' well at the same pumping rate with a temperature of 40 °C. This scenario is a hypothetical geothermal operation, but the geological site used is real and represents the typical geological complexity encountered during such an operation. It is thus a good test case for the methods developed and presented above, and used here to test and validate the workflow.

Mathematical model

The numerical simulation of the geothermal reservoir is carried out by the OGS simulator, which employs a fully coupled model of thermal and hydraulic processes in porous media (www.opengeosys.org). The governing equations of the relevant processes implemented in OGS are summarized in the following.

Fully saturated groundwater flow is expressed using a pressure-based formulation (Bear and Bachmat 1990):

$$S_0 \frac{\partial p_w}{\partial t} - \nabla \cdot \left(\frac{\mathbf{K}}{\mu_w} (\nabla p_w - \rho_w \cdot \mathbf{g}) \right) = Q_w \quad (1)$$

where S_0 (Pa⁻¹) is the storage coefficient, p_w (Pa) is the water pressure, t (s) is the time, ρ_w (kg/m³) is the water density, and μ_w (Pa·s) is the dynamic viscosity of water. $\mathbf{K}$ (m²) is the intrinsic permeability tensor, $\mathbf{g}$ (m/s²) is the gravitational acceleration and Q_w (1/s) is the source/sink term.

Heat transport in a fully water-saturated porous formation is governed by heat convection and conduction (Bear and Bachmat 1990):

$$c\rho \frac{\partial T}{\partial t} + \nabla \cdot (nc_w \rho_w \mathbf{v} T) - \nabla \cdot (\mathbf{D}_H \nabla T) = Q_T \quad (2)$$

where c and c_w [J/(kg·K)] are the specific heat capacities of the porous medium and of water, respectively. ρ (kg/m³) is the density of the porous medium, T (K) is the temperature, $\mathbf{v}$ (m/s) is the water flow velocity vector, n (–) is the porosity, $\mathbf{D}_H$ [W/(m·K)] is the heat conduction–dispersion tensor, and Q_T [J/(m³·s)] is a heat source/sink term.

In the presence of water in the porous medium, the volumetric heat capacity $c\rho$ in Eq. 2 is quantified here as

$$c\rho = nc_w \rho_w + (1 - n)c_s \rho_s \quad (3)$$

where c_s [J/(kg·K)] and ρ_s (kg/m³) are the specific heat capacity and density of the solid matrix, respectively.

The heat dispersion tensor $\mathbf{D}_H$ in Eq. 2 is formulated using volume averaging as

$$\mathbf{D}_H = n\lambda_w + (1-n)\lambda_s + nc_w\rho_w \left(\gamma_t |v| \delta_{ij} + (\gamma_l - \gamma_t) \frac{v_i v_j}{|v|} \right) \quad (4)$$

where λ_w and λ_s [W/(m·K)] are the thermal conductivities of water and porous medium, respectively, γ_l and γ_t (m) are the longitudinal and the transverse heat dispersivities. v_i and v_j (m/s) are the i and j components of the fluid velocity. δ_{ij} (—) is the Kronecker Delta.

The above coupled thermal and hydraulic processes are solved by a sequential iterative approach in OGS, where the standard Galerkin method is used to spatially discretize the weak forms and an implicit time stepping scheme for time discretization (Wang et al. 2009).

Reservoir model characterization

Subsurface geological model

The study site is located near Hamburg, northern Germany, and has a dimension of 17 km × 17 km width and 7 km depth. Sediment ages range from the Rotliegend up to Quaternary, forming a total of 20 geological formations (Fig. 5a). Due to the halokinesis of Rotliegend and Zechstein groups, the upraised salt forms a salt dome and intrudes into the overlying sediments from lower Buntsandstein up to the Oligocene. Above the salt dome, there exists a fault system from the Miocene base to the Quaternary base (Baldschuhn et al. 2001).

A static geological model is constructed in Petrel based on geological information on the near surface layers (from Miocene base to Quaternary base) from Scheer et al. (2007), and information on the deep layers (from Zechstein base to Oligocene base) from Hese (2011, 2012), who based his work on the geotectonic atlas of northern Germany (Baldschuhn et al. 2001). Vertical key pillars are used in Petrel to model the geometry of the salt dome and thus form the fault planes around the salt dome. This is because the fault plane is the only geometrical plane which can be vertical or inclined in the pillar-gridding approach to represent the geometry of the salt dome walls. As recommended in Schlumberger (2013), faults are modeled as ‘zigzag’ type. This helps to maintain the block faces in x - and y -directions as orthogonal as possible. The geological layers intruded by the salt dome are then intersected by the faults and thus show discontinuities across the salt dome. As a result, the salt dome is modeled as a gap in the geological model (Fig. 5a). For hydraulic process simulation, this way of geological modeling represents the true nature of the salt dome, which has an extremely low permeability because of highly dense compaction and thus reacts as a barrier for fluids. For heat flow simulations, the

salt dome should be modeled as a salt body with a given heat conductivity. Considering the geothermal reservoir in the application example, which is far away from the salt dome, we felt that this approximation is acceptable. For this reason and in order to reduce the computational effort, only the western part of the model up to the salt dome is used for the simulation of the geothermal reservoir operation.

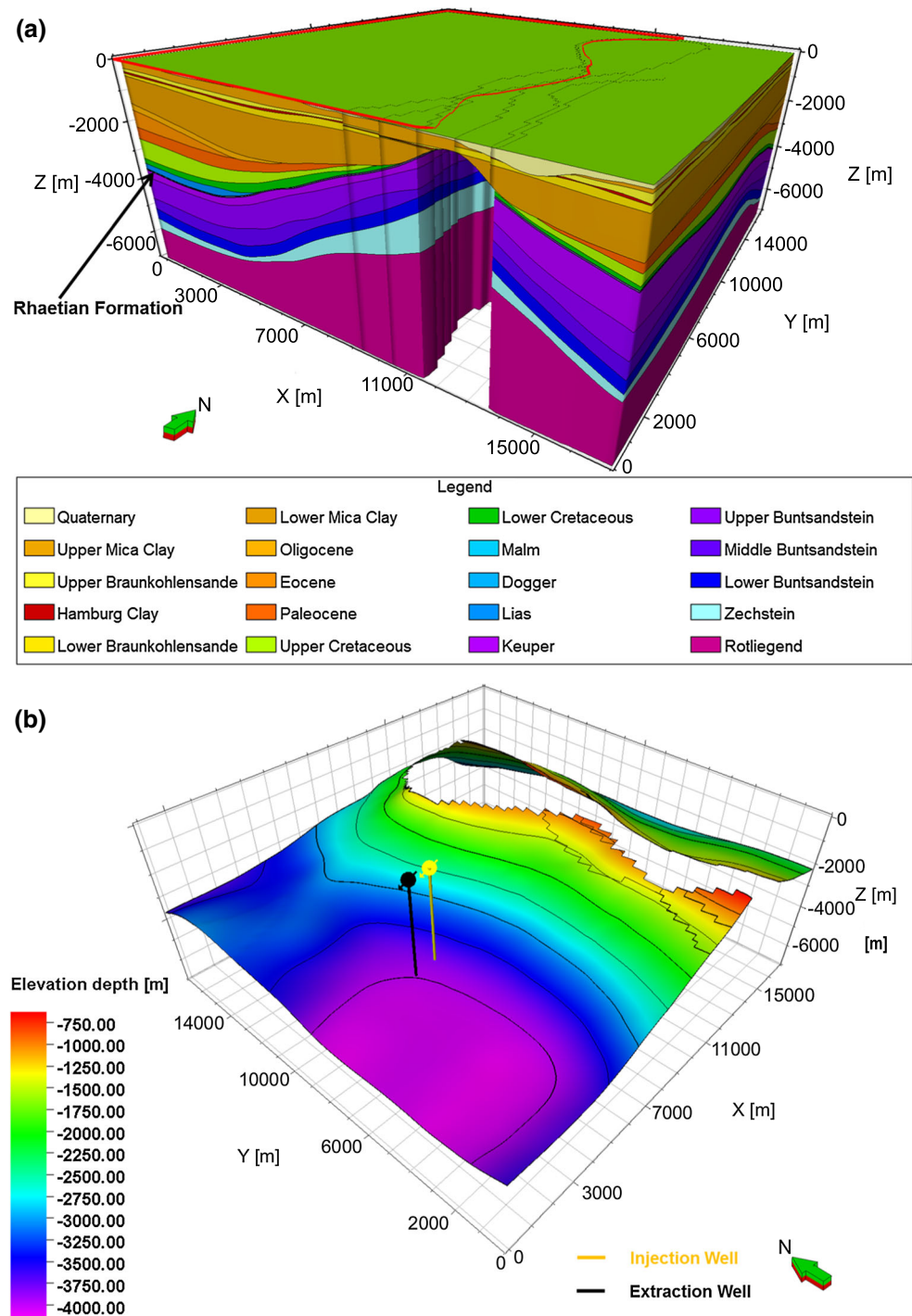
According to Reinhold et al. (2011) and Doornenbal and Stevenson (2010), the Rhaetian formation at the upper Triassic in northern Germany is a potential reservoir formation with porous sandstone and barrier shale layers. At our study site, the Rhaetian formation has a depth of about 4000 m at the west flank, which is deep enough to provide a high geothermal potential. Similar to the work of Motaghay et al. (2010) and Vogt et al. (2013), we thus choose the west Rhaetian formation as target formation. Based on the facies description from Hese (2011, 2012), the formation is further subdivided into three layers. The middle Rhaetian is the main sandstone layer with an average thickness of 20 m, and the upper and the lower Rhaetian are hydraulic barrier layers with an average thickness of 15 m each.

At the west side of the geological model, two vertical wells forming a well doublet are placed (Fig. 5b). The intersection of the extraction well with the middle Rhaetian formation is from 3963 to 3983 m depth. To minimize the possible temperature drop in the extraction well, the injection well is placed at 900 m distance, and with an intersection from 3803 to 3823 m depth. The wells are defined in Petrel and converted to the computational mesh using the newly developed methods explained above. In OpenGeoSys, the wells are geometrically represented by picked nodes, on which point sources are defined.

Model discretization and conversion

For stability and accuracy of the numerical simulation, the geological model within the simulated area is further refined. Along the z -direction, the middle Rhaetian formation is divided into ten layers with a thickness of 2 m each, and the upper and lower Rhaetian into five layers (2 × 5 m, 2 × 2 m and 1 × 1 m thickness). Near the middle Rhaetian, the layers have a thickness of 1 m. The other formations are also refined to get an average cell thickness of 100 m in each layer. Horizontally, the area around the well doublet is gradually refined, which means that the model domain within a distance of 150 m from the wells and between the two wells has a resolution of 12.5 m × 12.5 m, from 300 to 150 m distance the resolution is 25 m × 25 m, from 550 to 300 m distance the resolution is 50 m × 50 m, and from 1050 to 550 m the resolution is 100 m × 100 m. Beyond this, the model

Fig. 5 Subsurface static geological model. **a** Structural model including 20 geological formations with the *red line* indicating the extent of the numerical dynamic model region; **b** The Rhaetian formation layer and well doublet location. (Green arrows point north)



domain maintains a horizontal resolution of $200 \text{ m} \times 200 \text{ m}$ as in the original pillar grid.

As a result, the corner point grid of the geological model is discretized into $176 \times 123 \times 191$ blocks (i.e., 4,134,768 blocks in total). Among them, 2,587,462 blocks ($\sim 62.6 \%$) are active. Each formation in the model is assigned a unique material group index (Table 1). For a corner point grid-based simulator, e.g., Eclipse, which uses a finite difference scheme, the number of degrees of

freedom is equivalent to the number of blocks. This corner point grid is then exported from Petrel and converted using the newly developed mesh converter. The conversion is performed successfully, as all occurring geological features are correctly accounted for (see below). The converted hybrid FE mesh (Fig. 6a) has 3,340,336 elements in total, which is about 56.8 % more than the active blocks in the corner point grid, and it has 2,619,309 nodes, which means the number of degrees of freedom is 1.2 % higher for the

Table 1 Reservoir hydraulic and thermal properties (Mottaghy et al. 2010; Hese 2012; Stober and Bucher 2012)

Geological formation	Material group index	K (mD)	n (–)	S_0 (Pa ⁻¹)	C_s [KJ/(m ³ ·K)]	λ_s [W/(m·K)]	H (μW/m ³)
Quaternary	0	5000	0.40	4.70×10^{-10}	2100	2.5	0.9
Upper Mica Clay	1	0.1	0.30	4.65×10^{-10}	2500	2.5	0.9
Upper Braunkohlensande	2	5000	0.35	4.68×10^{-10}	2100	2.5	0.9
Hamburg Clay	3	0.1	0.30	4.65×10^{-10}	2500	2.5	0.9
Lower Braunkohlensande	4	1000	0.35	4.68×10^{-10}	2100	2.5	0.9
Lower Mica Clay	5	0.85	0.30	4.65×10^{-10}	2500	2.5	0.9
Oligocene	6	0.1	0.20	4.60×10^{-10}	2500	2.5	0.9
Eocene	7	0.1	0.20	4.60×10^{-10}	2500	2.5	0.9
Paleocene	8	0.1	0.20	4.60×10^{-10}	2500	2.5	0.9
Upper Cretaceous	9	1	0.20	4.60×10^{-10}	2300	2.9	0.9
Lower Cretaceous	10	10	0.20	4.60×10^{-10}	3000	2.0	0.9
Malm	11	0.1	0.15	4.58×10^{-10}	2300	1.2	1.4
Dogger	12	500	0.15	4.58×10^{-10}	2400	1.2	1.4
Lias	13	0.1	0.15	4.58×10^{-10}	2000	1.5	1.4
Upper Rhaetian	14	0.1	0.15	4.58×10^{-10}	2000	2.0	1.0
Middle Rhaetian	20	500	0.30	4.65×10^{-10}	2200	2.2	0.5
Lower Rhaetian	14	0.1	0.15	4.58×10^{-10}	2000	2.0	1.0
Keuper	14	0.1	0.15	4.58×10^{-10}	2000	2.0	1.0
Upper Buntsandstein	15	0.1	0.15	4.58×10^{-10}	1000	2.4	0.7
Middle Buntsandstein	16	10	0.15	4.58×10^{-10}	2200	2.4	0.7
Lower Buntsandstein	17	0.1	0.10	4.55×10^{-10}	1000	2.8	1.3
Zechstein	18	0.1	0.10	4.55×10^{-10}	840	4.5	0.5
Rotliegend	19	0.1	0.10	4.55×10^{-10}	1000	2.8	1.5

H (μW/m³) represents the natural heat generation from reservoir rocks

FE case. The FE mesh consists of 1,818,245 hexahedrons (~54.4 %), 1,464,905 prisms (~43.9 %), 24,306 pyramids (~0.7 %), and 32,880 tetrahedrons (~1.0 %). As a comparison, the conversion approach by Aarnes et al. (2008) in total yields 12,654,461 tetrahedrons elements, which is almost four times more than the new approach. By the new approach, the converted elements are mainly hexahedrons and prisms, which helps to maintain a reasonably small number of elements. In FE models, the node number indicates the computational burden when solving the equation system, while the element number is the burden for constructing the equation system and assembling the system matrix. Thus, this reduced element number enables faster dynamic model simulations and also requires less computer storage. The converted Rhaetian formation layer in combination with the well path is shown in Fig. 6b.

A closer view of the corner point grid and the converted mesh is shown in Fig. 6c, d in order to illustrate how the implemented conversion schemes have been applied to the geological model. Here, the upper formation is intruded by the lower one and also eroded. Block A in Fig. 6c has only

one zero-length edge in the lower right corner and is converted to prism A_1 and pyramid A_2 . In block B, two edges have zero length, and it is considered as a prism firstly. However, in order to keep consistent faces along the z-direction, it is further cut to one tetrahedron B_1 and one pyramid B_2 . Block C has three zero-length edges and is converted to two tetrahedrons. Because one tetrahedron inherits the three zero-length edges, it has zero volume and is removed from the mesh. Thus, block C is converted to only one tetrahedron C_1 . As a summary, about 29.7 % of the blocks in the corner point grid are partitioned by the implemented scheme, due to either the existence of degenerations or inconsistencies in joint faces. Therefore, our implemented methods convert the grid with degenerated blocks into the FE mesh with consistent hybrid elements, so the FE mesh is numerically healthy compared with the non-consistent direct conversion approaches. The results (see section ‘Simulation results’ below) from field application of 20 years’ geothermal reservoir operation show that we do not encounter numerical stability problems, which means that this conversion does not generate numerical instabilities.

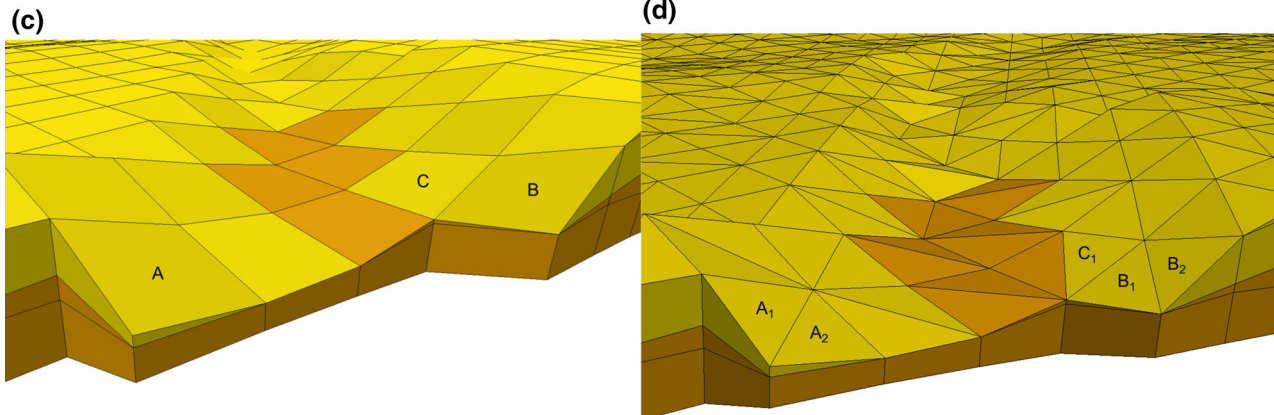
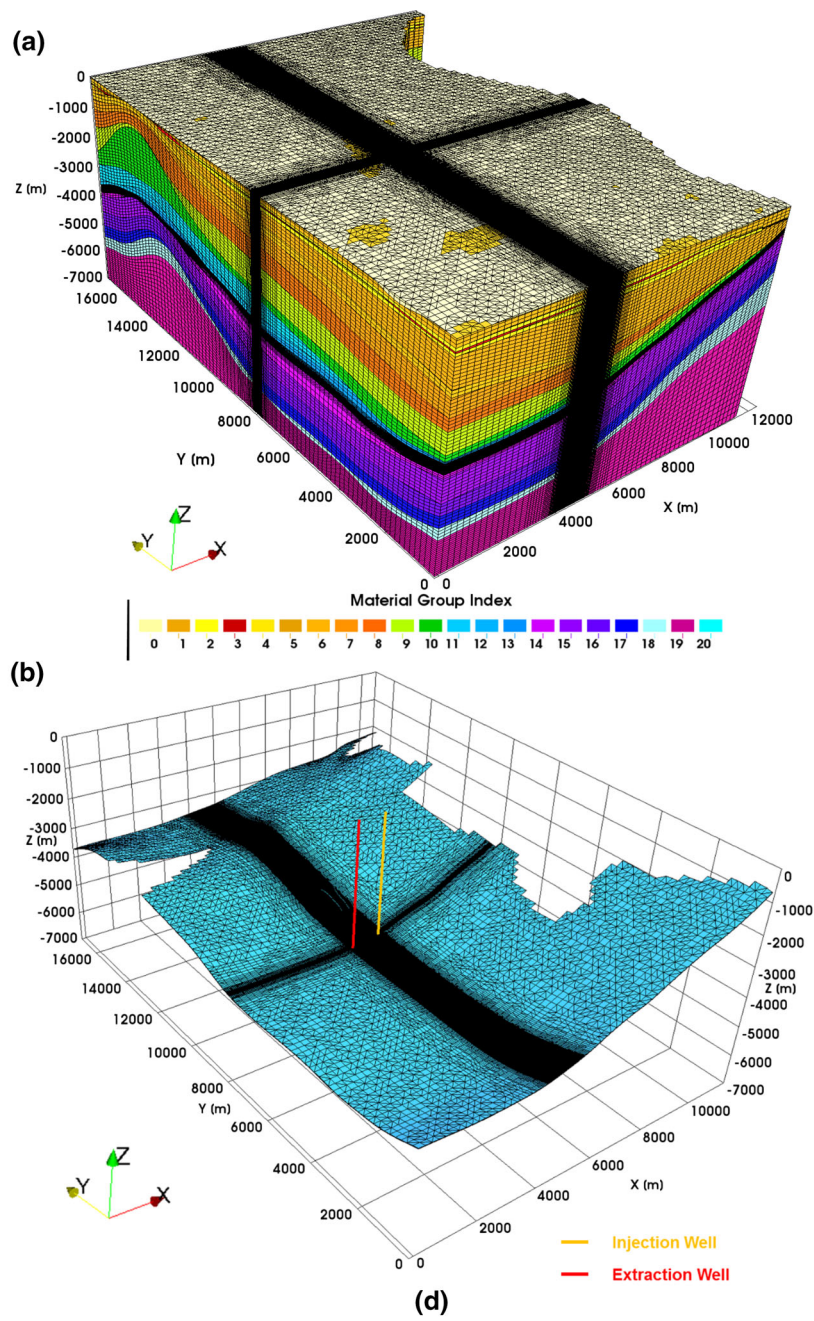


Fig. 6 Mesh conversion result. **a** Converted mesh with 20 material groups; **b** Converted mesh of the middle Rhaetian formation with two wells; **c** and **d** a closer comparison between the corner point *grid* and the converted mesh

Hydraulic and thermal properties

Non-isothermal flow is simulated in the model, and thus, the water dynamic viscosity is considered temperature dependent following the simple empirical model presented by Yaws and Gomes (2009). Also, the water density is defined by a linear temperature-dependent function according to Busch et al. (1993), in which the reference water density of 1000 kg/m^3 is set at reference temperature of 283.15 K and the thermal expansion coefficient to $-5 \times 10^{-4} \text{ K}^{-1}$ referring to Diersch and Kolditz (1998). Salinity effects are not considered here, as salt content in the formation is assumed homogenous. The other hydraulic and thermal properties (Table 1) are assumed homogenous in each geological formation. Regarding the thermal properties, the volumetric heat capacity is taken from Stober and Bucher (2012), and the thermal conductivity and natural heat production rate are taken from Mottaghy et al. (2010). The intrinsic permeability and porosity are taken from Scheer (2001), Doornenbal and Stevenson (2010), and Hese (2012). The storage coefficient is estimated based on the empirical function (Freeze and Cherry 1979) related to the compressibility of porous medium and water, which are assumed as $4.5 \times 10^{-10} \text{ Pa}^{-1}$ and $5 \times 10^{-10} \text{ Pa}^{-1}$, respectively.

Simulation results

In preparation for simulating the geothermal production, the initial reservoir conditions (i.e., initial pressure and temperature fields) have to be determined by two model runs. First, a stationary heat transport model is initialized with a temperature gradient of $30 \text{ }^\circ\text{C/km}$, and at the top surface ($z = 0 \text{ m}$), the temperature is set constant to $10 \text{ }^\circ\text{C}$ representing the mean annual air temperature. A constant heat flux of 70.5 mW/m^2 (Mottaghy et al. 2010) is applied at the bottom at $z = -7000 \text{ m}$, and natural heat generation is accounted for (Table 1). The resulting temperature field is taken as the initial temperature field for the stationary flow model in the second step. A pressure of 1 bar is set constant at the top as boundary condition, and the pressure field is initialized with a gradient of 9810 Pa/m . The resulting temperature versus depth profile corresponds well with the measured temperature profiles from Mottaghy et al. (2010), which are from the same area. This shows that the parametrization of the model is realistic, and temperatures correspond to real temperatures at respective depths (Fuchs and Förster 2010; Agemar et al. 2012; Suchi et al. 2014). The temperature in the reservoir formation at the

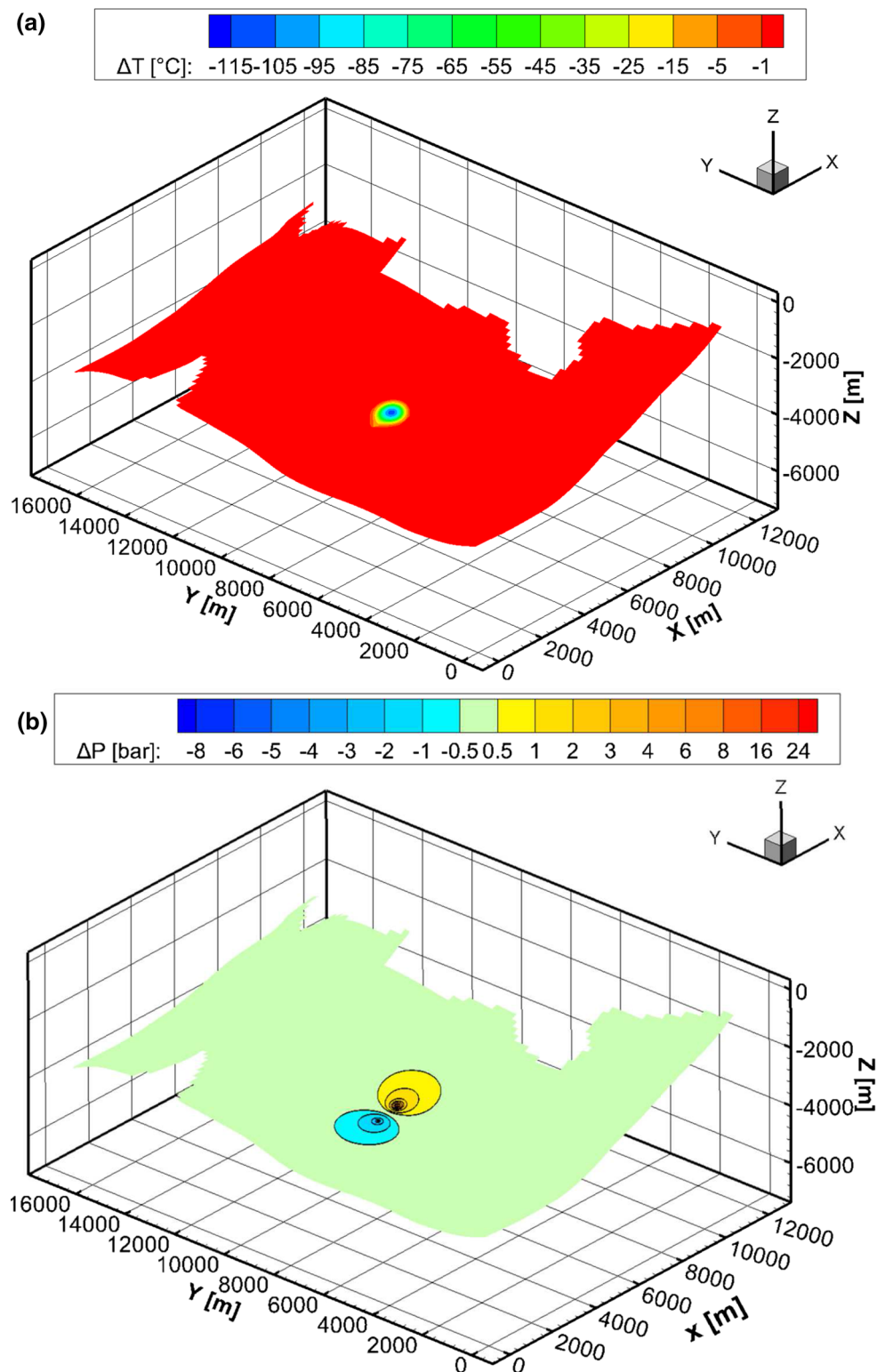
extraction well in 3980 m depth is $162.4 \text{ }^\circ\text{C}$, in 3000 m depth $116.2 \text{ }^\circ\text{C}$. These temperatures show that a geothermal use could be feasible.

The pressure and temperature fields described above are used as the initial conditions for the transient model. We add another Dirichlet boundary condition for the heat transport process, which is a constant temperature of $40 \text{ }^\circ\text{C}$ along the injection well in the middle Rhaetian to represent the temperature of injected water. This boundary condition is only applied when water is injected. For simulating pressure, the model boundary at the side of the salt dome is considered a no flow boundary, and the other three sides are open boundaries which maintain the initial water pressure. Due to the low permeability in the upper and lower Rhaetian formations, these are barrier formations and can prevent the water being displaced to neighboring layers. Accordingly, water flow is simulated only for the middle Rhaetian formation, while heat transport is simulated in the whole model domain. For the injection well, an injection rate of $150 \text{ m}^3/\text{h}$ is distributed to the nodes along the open section of the well path generated using the new converter, and at the extraction well, a rate of $-150 \text{ m}^3/\text{h}$ is assigned accordingly. The total simulation time is set to 20 years, and within each year, the well doublet operates only for six consecutive months.

We simulated the complete model as shown in Fig. 6a. Because our application interest, i.e., the geothermal reservoir, is the porous formation in the sandstone layer of the middle Rhaetian, the model results here only show the temperature and pressure changes in the Rhaetian formation. The temperature perturbation in the Rhaetian formation after 20 years is shown in Fig. 7a. A temperature decrease of $1 \text{ }^\circ\text{C}$ or larger is observed within 900 m of the injection well showing a radial transport of injected water and heat away from the injection well. At the injection well, temperature drops by $111 \text{ }^\circ\text{C}$ to $44 \text{ }^\circ\text{C}$. Figure 7b shows the pressure response at the end of the six-month operation period in the last year of the simulation. Pressure changes of up to 24 bars and eight bars are observed at the bottom of the injection and the extraction wells, respectively. At roughly 1.5 km away from the center of both wells, pressure response is 1 bar. In contrast to the temperature distribution, the pressure has a larger region of impact and the pressure change caused by the injection is larger than the pressure change caused by the extraction. This is due to reservoir cooling near the injection well, as water viscosity is increased there and thus increases flow resistance (see Eq. 1), which leads to a larger pressure change.

A closer view of the temperature changes after 20 years at the well doublet is shown in Fig. 8. The cold-water front propagates away from the injection well (Fig. 8a) and preferentially moves toward the extraction well following the steepest hydrological gradient. At the injection well,

Fig. 7 **a** Temperature changes in the middle Rhaetian formation after 20 years; **b** Pressure changes after the injection in the last cycle



temperature drops to 44 °C and leads to a temperature decrease of 111 °C. The temperature perturbation in the x - y plane (Fig. 8b) shows a temperature decrease of 1 °C at a distance of about 900 m away from the injection well. In the x - z plane (Fig. 8c), the temperature also changes above

and below the reservoir formation due to heat conduction, and a decrease of 1 °C is observed at a distance of roughly 90 m. The temperature profile at the bottom of the extraction well is recorded over 20 years (Fig. 8d). The overall temperature decrease is 1.5 °C, and a relatively

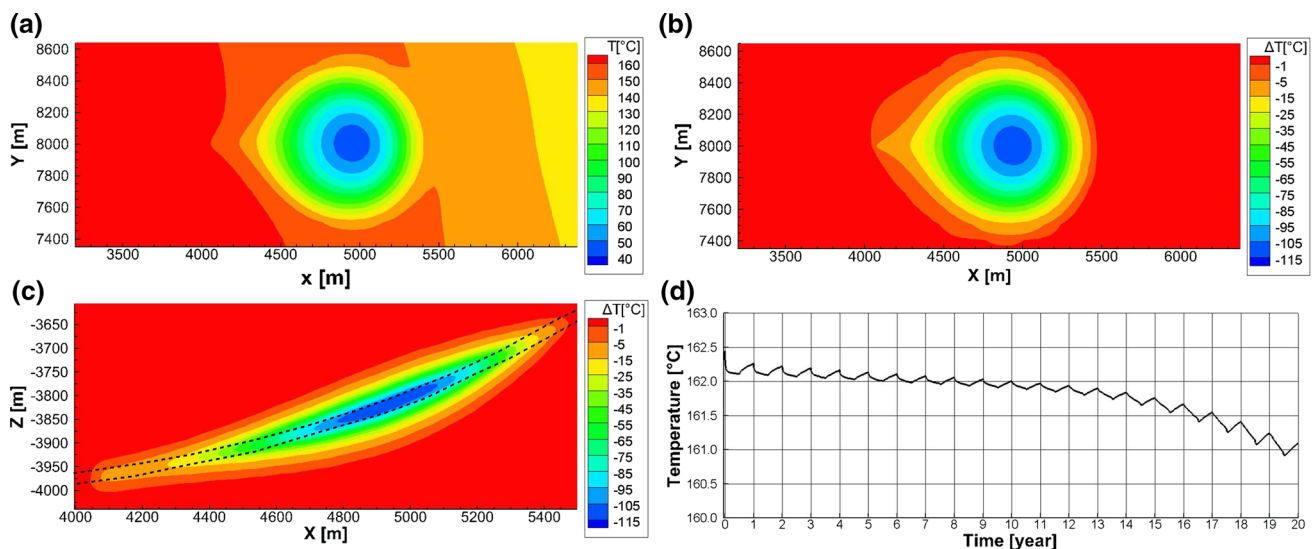


Fig. 8 **a** Temperature distribution near the well doublet in the x - y plane; **b** Temperature change near the well doublet in the x - y and the x - z planes (**c**), where the *black dashed line* indicates the reservoir formation; **d** Temperature profile at the *bottom* of the extraction well

faster drop of 1 °C is within the last 8 years, indicating interference with the advancing cold-water front. Overall, water with an average temperature of 161.7 °C at a rate of 150 m³/h is produced for 6 months in each year. This corresponds to an energy production rate of 21.2 MW, considering a 40 °C reinjection temperature. Over 20 years, this corresponds to 1854 GWh of produced thermal energy.

Summary and conclusions

In order to investigate possible uses of the geological subsurface and their potential impacts, scenario simulations have to be conducted on the basis of complex static geological models representing the real geological system. We developed a workflow to transfer complex heterogeneous geological models to consistent FE models. Three functionalities have been implemented. First, a corner point grid is consistently converted accounting for the individual block situation, and providing a hybrid FE mesh containing elements of hexahedron, prism, pyramid, and tetrahedron type. Secondly, the heterogeneous properties of the corner point grid are transferred from individual blocks to corresponding elements. Third, well trajectories can be mapped to the nodes or later formed as polylines in the FE mesh and used as source terms. The methods were tested extensively and verified in a series of technical test cases.

Instead of remeshing or reconstructing approaches (e.g., Blöcher et al. 2010a; Zehner et al. 2015), our implemented methods can be straightforwardly applied to convert existing corner point grids based on geological models to FE meshes for dynamic models. Besides, this direct

conversion from blocks to elements allows for consistency when transferring spatially heterogeneous material properties, which avoids to reparametrize the converted FE models. This approach additionally provides the possibility to couple the commercial Eclipse simulator with OpenGeoSys (Pfeiffer et al. 2016) and especially to simulate induced geochemical impacts in the use of subsurface. Among the existing direct conversion approaches (e.g., Graupner et al. 2011; Ni and Chen 2014; Park et al. 2014), which handle only hexahedral blocks, the method presented here accounts for all kinds of degenerated blocks and allows to construct a consistent FE mesh. In order to reduce element numbers in converted FE mesh and achieve a higher efficiency in running time, we choose to use all FE element types instead of only tetrahedrons (Aarnes et al. 2008).

Although the direct conversion methods presented in this manuscript are comparatively fast, they come with some drawbacks because the geological models cannot be changed or modified during the conversion. The discretization and parameterization of the geological model must be finalized before the conversion, already including all requirements from the process simulation like the grid refinement near the injection and extraction wells in the demonstrated application due to numerical reasons. If, for example, during process simulations, it is found that a finer mesh is required due to numerical reasons, the mesh has to be changed in the geological model and the conversion process repeated. However, only the mesh has to be adapted, not the geological structure or parametrization. This repeated conversion is why an automated and fast conversion method is required. The quality of the converted mesh highly depends on the quality of the input

corner point grid, which should be checked for, e.g., zero-volume elements, using functionality provided by the meshing tool Petrel after discretizing the geological models. Comparing this method to the remeshing or the reconstructing approach, the representation of geological complexities in the converted mesh is restricted by the possibilities of representation in the original corner point grid. For example, subgrid fracture networks cannot be modeled geometrically but only as grid properties and thus cannot be integrated explicitly in the converted mesh. Besides, although the non-neighbor blocks at two sides of the fault slip is consistently converted to the hexahedral elements, the fault slip appears inevitably as hanging nodes in the converted mesh, which might not be compatible with some FE simulators. If these features are of importance, the alternative methods have to be employed.

To demonstrate the application of the implemented workflow, we performed a scenario study on a geothermal reservoir operation in northern Germany. A static geological model was built for the study site using Petrel. The resulting model comprises 20 geological formations of different materials combined with one salt dome structure and several faults, which are discretized into 2,587,462 active blocks in its corner point grid. For the grid conversion to an FE mesh, about 29.7 % of the grid blocks needed to be partitioned due to the existence of degenerations. The conversion resulted in a hybrid FE mesh with 3,340,336 elements, including 54.4 % of hexahedrons, 43.9 % of prisms, 0.7 % of pyramids, and 1.0 % of tetrahedrons. The mesh has roughly a quarter of the elements compared with the method of Aarnes et al. (2008), which is the only consistent method given in literature so far. This reduced element number enables faster dynamic model simulations and also requires less computer storage. In order to differentiate geological formations in the FE mesh, a material index has been transferred as property data. Also, the trajectories of a well doublet were mapped from Petrel to the nodes of the FE mesh. This allows for an accurate placement of the wells, consistent with the geological model. The converted FE model was then simulated by OGS in a coupled heat transport and water flow model. The simulation demonstrates that the converted FE mesh can be employed for the required dynamic simulation, allowing for stable numerical results. The dynamic model can be used to dimension the heat retrieval well doublet system and give estimates of induced effects.

Besides this application shown here, the presented methods can also be applied, for instance, to investigate oil and gas recovery, underground storage of natural gas, hydrogen and compressed air, and brine disposal in deep saline porous formations. Although Petrel, for which the conversion methods were implemented in the work presented here, is a commercial tool, it is widely used and

provides suitable functionality for representing subsurface structures, which can by this method be made use of for numerical process simulations. However, the presented methods are not restricted to operate only with this software for OGS, but, with straightforward modifications of the input and output functions, they can be generally employed for converting any type of corner point grid-based geological models to any FE models format for other geological modeling and numerical simulation codes.

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References

- Aarnes JE, Krogstad S, Lie K-A (2008) Multiscale mixed/mimetic methods on corner-point grids. *Comput Geosci* 12:297–315. doi:[10.1007/s10596-007-9072-8](https://doi.org/10.1007/s10596-007-9072-8)
- Agemar T, Schellschmidt R, Schulz R (2012) Subsurface temperature distribution in Germany. *Geothermics* 44:65–77. doi:[10.1016/j.geothermics.2012.07.002](https://doi.org/10.1016/j.geothermics.2012.07.002)
- Baldschuhn R, Binot F, Fleig S, Kockel F (2001) Geotektonischer Atlas von Nordwest-Deutschland und dem-deutschen Nordsee-Sektor. Schweizerbart Science Publishers, Stuttgart
- Bauer S, Beyer C, Dethlefsen F, Dietrich P, Duttman R, Ebert M, Feeser V, Görke U, Köber R, Kolditz O, Rabbel W, Schanz T, Schäfer D, Würdemann H, Dahmke A (2013) Impacts of the use of the geological subsurface for energy storage: an investigation concept. *Environ Earth Sci* 70:3935–3943. doi:[10.1007/s12665-013-2883-0](https://doi.org/10.1007/s12665-013-2883-0)
- Bauer S, Pfeiffer T, Boockmeyer A, Dahmke A, Beyer C (2015) Quantifying induced effects of subsurface renewable energy storage. *Energy Procedia* 76:633–641. doi:[10.1016/j.egypro.2015.07.885](https://doi.org/10.1016/j.egypro.2015.07.885)
- Bear J, Bachmat Y (1990) Introduction to modeling of transport phenomena in porous media. Kluwer, AA Dordrecht
- Benisch K, Bauer S (2013) Short- and long-term regional pressure build-up during CO₂ injection and its applicability for site monitoring. *Int J Greenh Gas Control* 19:220–233. doi:[10.1016/j.ijggc.2013.09.002](https://doi.org/10.1016/j.ijggc.2013.09.002)
- Benisch K, Köhn D, Al Hagrey S, Rabbel W, Bauer S (2014) A combined seismic and geoelectrical monitoring approach for CO₂ storage using a synthetic field site. *Environ Earth Sci* 73:3077–3094. doi:[10.1007/s12665-014-3603-0](https://doi.org/10.1007/s12665-014-3603-0)
- Blöcher MG, Cacace M, Lewerenz B, Zimmermann G (2010a) Three dimensional modelling of fractured and faulted reservoirs: framework and implementation. *Chemie der Erde Geochem* 70:145–153. doi:[10.1016/j.chemer.2010.05.014](https://doi.org/10.1016/j.chemer.2010.05.014)
- Blöcher MG, Zimmermann G, Moeck I, Brandt W, Hassanzadegan A, Magri F (2010b) 3D numerical modeling of hydrothermal processes during the lifetime of a deep geothermal reservoir. *Geofluids* 10:406–421. doi:[10.1111/j.1468-8123.2010.00284.x](https://doi.org/10.1111/j.1468-8123.2010.00284.x)
- Boockmeyer A, Bauer S (2014) High-temperature heat storage in geological media: high-resolution simulation of near-borehole

- processes. *Géotech Lett* 4:151–156. doi:[10.1680/geolett.13.00060](https://doi.org/10.1680/geolett.13.00060)
- Busch K, Luckner L, Tiemer K (1993) *Geohydraulik, Lehrbuch der Hydrogeologie*. Gebrüder Borntraeger, Berlin
- Cacace M, Blöcher G (2015) MeshIt—a software for three dimensional volumetric meshing of complex faulted reservoirs. *Environ Earth Sci* 74:5191–5209. doi:[10.1007/s12665-015-4537-x](https://doi.org/10.1007/s12665-015-4537-x)
- Diersch H-JG, Kolditz O (1998) Coupled groundwater flow and transport: 2. Thermohaline and 3D convection systems. *Adv Water Resour* 21:401–425. doi:[10.1016/S0309-1708\(97\)00003-1](https://doi.org/10.1016/S0309-1708(97)00003-1)
- Doornenbal H, Stevenson A (2010) *Petroleum geological atlas of the Southern Permian Basin area*. EAGE, Houten, the Netherlands
- Fischer T, Naumov D, Sattler S, Kolditz O, Walther M (2015) GO2OGS 1.0: a versatile workflow to integrate complex geological information with fault data into numerical simulation models. *Geosci Model Dev* 8:3681–3694. doi:[10.5194/gmd-8-3681-2015](https://doi.org/10.5194/gmd-8-3681-2015)
- Franco A, Vaccaro M (2014) Numerical simulation of geothermal reservoirs for the sustainable design of energy plants: a review. *Renew Sustain Energy Rev* 30:987–1002. doi:[10.1016/j.rser.2013.11.041](https://doi.org/10.1016/j.rser.2013.11.041)
- Freeze R, Cherry J (1979) *Groundwater*. Prentice-Hall Inc, New Jersey
- Fuchs S, Förster A (2010) Rock thermal conductivity of Mesozoic geothermal aquifers in the Northeast German Basin. *Chemie der Erde - Geochem* 70:13–22. doi:[10.1016/j.chemer.2010.05.010](https://doi.org/10.1016/j.chemer.2010.05.010)
- Graupner BJ, Li D, Bauer S (2011) The coupled simulator ECLIPSE–OpenGeoSys for the simulation of CO₂ storage in saline formations. *Energy Procedia* 4:3794–3800. doi:[10.1016/j.egypro.2011.02.314](https://doi.org/10.1016/j.egypro.2011.02.314)
- Hese F (2011) *Geologische 3D-Modelle des Untergrundes Schleswig-Holsteins—ein Beitrag für Potenzialstudien zur Nutzung von tiefen salinen Aquiferen*. Zeitschrift der Deutschen Gesellschaft für Geowissenschaften 162:389–404. doi:[10.1127/1860-1804/2011/0162-0389](https://doi.org/10.1127/1860-1804/2011/0162-0389)
- Hese F (2012) *3D Modellierungen und Visualisierung von Untergrundstrukturen für die Nutzung des unterirdischen Raumes in Schleswig-Holstein*. Ph.D. Thesis, University of Kiel, Germany
- Kalbacher T, Wang W, McDermott C, Kolditz O, Taniguchi T (2005) Development and application of a CAD interface for fractured rock. *Environ Geol* 47:1017–1027. doi:[10.1007/s00254-005-1236-z](https://doi.org/10.1007/s00254-005-1236-z)
- Kalbacher T, Mettler R, McDermott C, Wang W, Kosakowski G, Taniguchi T, Kolditz O (2007) Geometric modelling and object-oriented software concepts applied to a heterogeneous fractured network from the Grimsel rock laboratory. *Comput Geosci* 11:9–26. doi:[10.1007/s10596-006-9032-8](https://doi.org/10.1007/s10596-006-9032-8)
- Kolditz O, Bauer S (2004) A process-oriented approach to computing multi-field problems in porous media. *J Hydroinformatics* 6:225–244
- Li D, Bauer S, Benisch K, Graupner B, Beyer C (2014) OpenGeoSys-ChemApp: a coupled simulator for reactive transport in multi-phase systems and application to CO₂ storage formation in Northern Germany. *Acta Geotech* 9:67–79. doi:[10.1007/s11440-013-0234-7](https://doi.org/10.1007/s11440-013-0234-7)
- Miles B, Kalbacher T, Kolditz O, Chen C, Gronewold J, Wang W, Peter A (2007) Development and parameterisation of a complex hydrogeological model based on high-resolution direct-push data. *Environ Geol* 52:1399–1412. doi:[10.1007/s00254-006-0582-9](https://doi.org/10.1007/s00254-006-0582-9)
- Mitiku AB, Li D, Bauer S, Beyer C (2013) Geochemical modelling of CO₂–water–rock interactions in a potential storage formation of the North German sedimentary basin. *Appl Geochem* 36:168–186. doi:[10.1016/j.apgeochem.2013.06.008](https://doi.org/10.1016/j.apgeochem.2013.06.008)
- Mottaghy D, Pechnig R, Taugis R, Kröger J, Thomsen C, Hese F, Liebsch-Doerschner T (2010) Erstellung eines geothermischen Modells für Teile Hamburgs und anliegende Gebiete. *BBR Jahresmagazin* 52–59
- Neave JW (2007) Analysis and characterization of fault networks. Google Patents
- Ni XD, Chen K (2014) Study on the conversion of GOCAD models to FLAC3D models. *Appl Mech Mater* 501–504:2527–2531
- Pan L, Oldenburg CM, Pruess K, Wu Y-S (2011) Transient CO₂ leakage and injection in wellbore-reservoir systems for geologic carbon sequestration. *Greenh Gases Sci Technol* 1:335–350. doi:[10.1002/ghg](https://doi.org/10.1002/ghg)
- Paradigm (2016) Paradigm GOCAD. <http://www.pdgm.com/products/gocad/>. Accessed 22 Feb 2016
- Park CH, Shinn YJ, Park YC, Huh DG, Lee SK (2014) PET2OGS: algorithms to link the static model of Petrel with the dynamic model of OpenGeoSys. *Comput Geosci* 62:95–102. doi:[10.1016/j.cageo.2013.09.014](https://doi.org/10.1016/j.cageo.2013.09.014)
- Pfeiffer WT, Bauer S (2015) Subsurface porous media hydrogen storage—scenario development and simulation. *Energy Procedia* 76:565–572. doi:[10.1016/j.egypro.2015.07.872](https://doi.org/10.1016/j.egypro.2015.07.872)
- Pfeiffer WT, Graupner B, Bauer S (2016) The coupled non-isothermal, multiphase-multicomponent flow and reactive transport simulator OpenGeoSys-Eclipse for porous media gas storage. Accessed at http://angusplus.de/en/publications/journal-articles?set_language=en
- Ponting DK (1992) Corner point geometry in reservoir simulation. *Math Oil Recover Inst Math its Appl*. doi:[10.3997/2214-4609.201411305](https://doi.org/10.3997/2214-4609.201411305)
- Reinhold K, Müller C, Riesenberger C (2011) *Informationssystem Speichergesteine für den Standort Deutschland-Synthese*. Bundesanstalt für Geowissenschaften und Rohstoffe (BGR), Berlin
- Sa H, Kohn D, Ce W, Schafer D, Rabbel W (2014) Feasibility study for geophysical monitoring renewable gas energy compressed in pore storages. *J Geol Geosci*. doi:[10.4172/2329-6755.1000169](https://doi.org/10.4172/2329-6755.1000169)
- Scheer W (2001) *Untersuchungsprogramm zur Ermittlung des nutzbaren Grundwasserdargebotes im schleswig-holsteinischen Nachbarraum zu Hamburg, Südost-Holstein*. Abschlußbericht.Landesamt für Natur und Umwelt des Landes Schleswig-Holstein. Flintbek. 52 S
- Scheer W, Kröger J, Reinhard K (2007) *Geologische 3-D-Modellierung des untergrundes—Ergebnisse aus dem INTERREG IIIB-Projekt BurVal*. Jahresbericht LANU, Schleswig-Holstein
- Schlumberger (2013) 2013 Petrel structural modeling—training and Exercise Guide. Schlumberger
- Schlumberger (2014a) Petrel E&P Software Platform version 2014.6
- Schlumberger (2014b) Eclipse reservoir simulation software v2014.1—Technical Description Manual. Schlumberger Ltd
- Souche L, Lepage F, Iskenova G (2013) Volume based modeling—automated construction of complex structural models. In: 75th EAGE Conf Exhib Inc SPE Eur 2013 London, UK, 10–13 June 2013 10–13. doi:[10.3997/2214-4609.20130037](https://doi.org/10.3997/2214-4609.20130037)
- Stober I, Bucher K (2012) *Geothermie*. Springer, Berlin
- Suchi E, Dittmann J, Knopf S, Müller C, Schulz R (2014) *Geothermie-atlas zur Darstellung möglicher Nutzungskonkurrenzen zwischen CCS und Tiefer Geothermie in Deutschland*. Zeitschrift der Deutschen Gesellschaft für Geowissenschaften 165:439–453. doi:[10.1127/1860-1804/2014/0070](https://doi.org/10.1127/1860-1804/2014/0070)
- Vogt C, Iwanowski-Strahser K, Marquart G, Arnold J, Mottaghy D, Pechnig R, Gnjezda D, Clauser C (2013) Modeling contribution to risk assessment of thermal production power for geothermal reservoirs. *Renew Energy* 53:230–241. doi:[10.1016/j.renene.2012.11.026](https://doi.org/10.1016/j.renene.2012.11.026)
- Wang W, Kosakowski G, Kolditz O (2009) A parallel finite element scheme for thermo-hydro-mechanical (THM) coupled problems in porous media. *Comput Geosci* 35:1631–1641. doi:[10.1016/j.cageo.2008.07.007](https://doi.org/10.1016/j.cageo.2008.07.007)
- Watanabe N, Wang W, McDermott CI, Taniguchi T, Kolditz O (2009) Uncertainty analysis of thermo-hydro-mechanical

- coupled processes in heterogeneous porous media. *Comput Mech* 45:263–280. doi:[10.1007/s00466-009-0445-9](https://doi.org/10.1007/s00466-009-0445-9)
- Yaws CL, Gomes J (2009) Transport properties of chemicals and hydrocarbons. William Andrew, New York
- Zehner B, Börner JH, Görz I, Spitzer K (2015) Workflows for generating tetrahedral meshes for finite element simulations on complex geological structures. *Comput Geosci* 79:105–117. doi:[10.1016/j.cageo.2015.02.009](https://doi.org/10.1016/j.cageo.2015.02.009)